

Group Index of 450 nm and 2 μm TE Silicon Strip Waveguides from Unbalanced MZIs, and Ring-Resonator Loss

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Abstract

Design proposal for chip EBeam_USER2. Y-branch MZIs with 450×220 nm arms ($\Delta L = 25/50/75/100/200 \mu\text{m}$) and one MZI with $2 \mu\text{m}$ wide arms ($\Delta L = 100 \mu\text{m}$) give the TE0 group index (simulated $n_g = 4.312$ and 3.761 at 1550 nm). Five all-pass rings ($R = 10 \mu\text{m}$) give the propagation loss.

1 Introduction

n_g sets the FSR of rings and interferometers and the length of phase shifters. Wide waveguides are used for low-loss routing and delay lines. Ring Q gives the propagation loss.

Objectives:

- $n_g(\lambda)$ of TE0, $w = 450$ nm, from the MZI FSR over five ΔL ;
- n_g of TE0, $w = 2 \mu\text{m}$, from one MZI;
- propagation loss from the intrinsic Q of rings with five gaps.

2 Theory

2.1 Waveguide Compact Model

$$n_{\text{eff}}(\lambda) = n_1 + n_2(\lambda - \lambda_0) + n_3(\lambda - \lambda_0)^2, \quad n_g = n_{\text{eff}} - \lambda \frac{dn_{\text{eff}}}{d\lambda}. \quad (1)$$

2.2 MZI Transfer Function

Two ideal Y-branches [1]:

$$\frac{I_o}{I_i} = \frac{1}{4} \left| e^{-j\beta_1 L_1 - \frac{\alpha}{2} L_1} + e^{-j\beta_2 L_2 - \frac{\alpha}{2} L_2} \right|^2, \quad \beta = \frac{2\pi n_{\text{eff}}(\lambda)}{\lambda}. \quad (2)$$

Identical, lossless arms, $\Delta L = L_2 - L_1$:

$$\frac{I_o}{I_i} = \frac{1}{2} [1 + \cos(\beta \Delta L)]. \quad (3)$$

Tapers are identical in both arms; ΔL lies entirely in the 450 nm (or 2 μm) sections.

2.3 Group Index from the FSR

Arm phase difference $\phi = 2\pi n_{\text{eff}} \Delta L / \lambda$. Adjacent minima differ by $\Delta\phi = 2\pi$, so $|d\phi/d\lambda| \text{ FSR} = 2\pi$ with

$$\frac{d\phi}{d\lambda} = -\frac{2\pi\Delta L}{\lambda^2} \left(n_{\text{eff}} - \lambda \frac{dn_{\text{eff}}}{d\lambda} \right) \Rightarrow \underset{\sim}{n_g} = \frac{\lambda^2}{\Delta L \cdot \text{FSR}}. \quad (4)$$

For several ΔL :

$$\frac{1}{\text{FSR}} = \frac{n_g}{\lambda^2} \Delta L, \quad \frac{\delta n_g}{n_g} \approx \sqrt{2} \frac{\delta \lambda_{\text{min}}}{\text{FSR}}. \quad (5)$$

2.4 Ring Resonator

All-pass ring, $r = \sqrt{1 - \kappa^2}$, $a = e^{-\alpha L/2}$, $\theta = \beta L$ [2]:

$$T = \frac{a^2 - 2\text{arccos}\theta + r^2}{1 - 2\text{arccos}\theta + (ar)^2}, \quad Q_i = \frac{2\pi n_g}{\lambda \alpha}. \quad (6)$$

Critical coupling at $r = a$.

3 Modelling and Simulation

3.1 Waveguides

Si strip, $h = 220$ nm, SiO_2 cladding, TE polarization; $w = 450$ nm (PDK type “Strip TE-TM 1550, w=450 nm”) and $w = 2 \mu\text{m}$ (multimode; TE0 launched by adiabatic 0.5→2 μm tapers) [3]. Lumerical MODE (FDE) [4]: Figs. 1, 2.

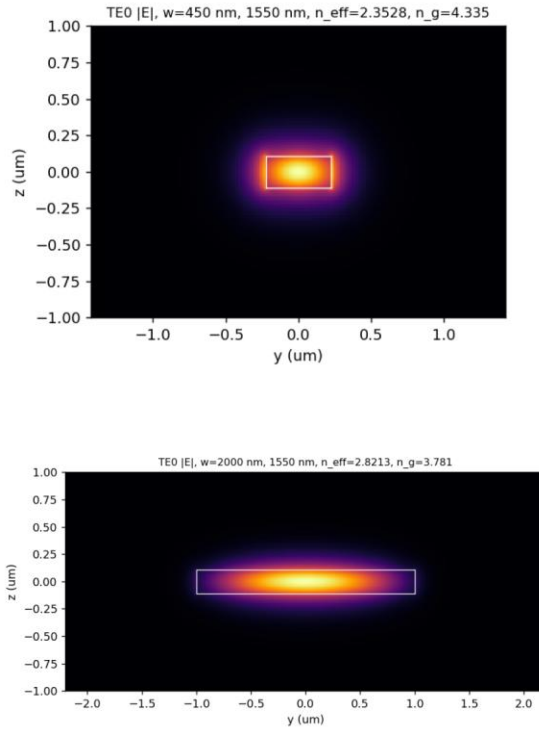


Fig. 1. TE0 mode profiles ($|E|$) at 1550 nm, MODE (FDE): $w = 450$ nm (top), $2 \mu\text{m}$ (bottom).

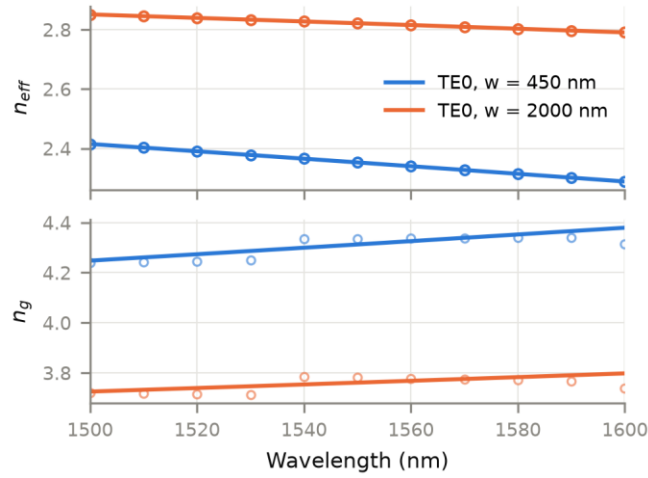


Fig. 2. n_{eff} and n_g vs. wavelength. Markers: MODE (FDE); lines: Eqs. (7), (8).

Compact models, λ in μm , $\lambda_0 = 1.55 \mu\text{m}$:

$$n_{\text{eff},450}(\lambda) = 2.3528 - 1.2643(\lambda - \lambda_0) - 0.4240(\lambda - \lambda_0)^2, \quad (7)$$

$$n_{\text{eff},2000}(\lambda) = 2.8213 - 0.6062(\lambda - \lambda_0) - 0.2343(\lambda - \lambda_0)^2. \quad (8)$$

$n_g(1550) = 4.312$ ($w = 450$ nm) and 3.761 ($w = 2$ μm).

3.2 MZI Design

TABLE 1. Expected FSR from Eqs. (7), (8)

Device	ΔL (μm)	FSR (nm) at $\lambda =$			pts/ FSR
		1500	1550	1600	
MZITEw450dL25	25	21.19	22.28	23.38	21188
MZITEw450dL50	50	10.59	11.14	11.69	10594
MZITEw450dL75	75	7.06	7.43	7.79	7063
MZITEw450dL100	100	5.30	5.57	5.85	5297
MZITEw450dL200	200	2.65	2.79	2.92	2648
MZITEw2000dL100	100	6.04	6.39	6.74	6040

ebeam_y_1550 split and combine. FSR < 50 nm and > 2600 points per FSR at 1 pm (Table 1).

3.3 Circuit Simulation

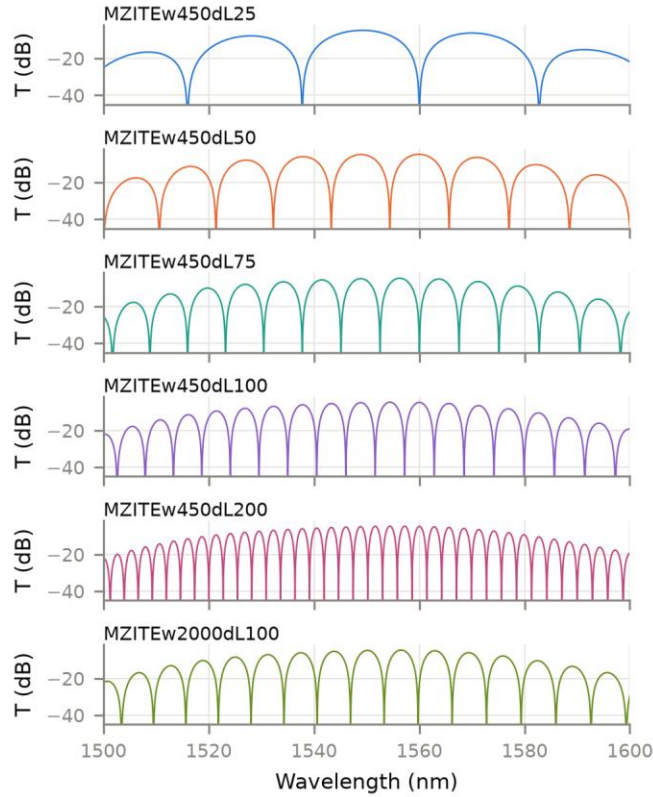


Fig. 3. Simulated MZI transmission, GC to GC: Eq. (3) with Eqs. (7), (8) and the PDK TE GC response [3].

Fig. 3.

3.4 Extraction Test on Simulated Spectra

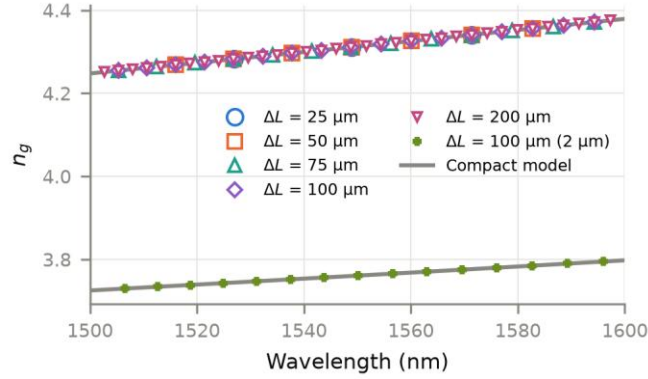


Fig. 4. n_g from the simulated FSR, Eq. (4) (markers), and compact models (lines).

TABLE 2. n_g at 1550 nm extracted from Fig. 3

Device	FSR (nm)	n_g
MZITEw450dL25	22.284	4.3126
MZITEw450dL50	11.142	4.3124
MZITEw450dL75	7.428	4.3124
MZITEw450dL100	5.571	4.3124
MZITEw450dL200	2.786	4.3124
Slope, Eq. (5), MZITEw450		4.3124
MZITEw2000dL100	6.388	3.7609

Eqs. (4), (5) recover the model n_g (Fig. 4, Table 2).

3.5 Procedure for Measured Data

1. Subtract the LBTE spectrum (dB).
2. Locate the minima; quadratic fit around each null (sub-pm).
3. FSR of adjacent minima $\rightarrow n_g(\lambda)$, Eq. (4).
4. Slope over ΔL , Eq. (5) ($w = 450$ nm); single-device value for $w = 2$ μm ; compare with the compact models.

3.6 Ring Resonators

TABLE 3. Expected ring response at 1550 nm ($R = 10$ μm , 3 dB/cm assumed)

Device	κ^2	FSR (nm)	Q_L	ER (dB)
RingTEg150	0.0308	9.77	28k	2.3

Device	κ^2	FSR (nm)	Q_L	ER (dB)
RingTEg200	0.0127	9.77	59k	5.7
RingTEg250	0.0052	9.77	108k	17.8
RingTEg300	0.0021	9.77	162k	10.3
RingTEg350	0.0009	9.77	203k	3.8

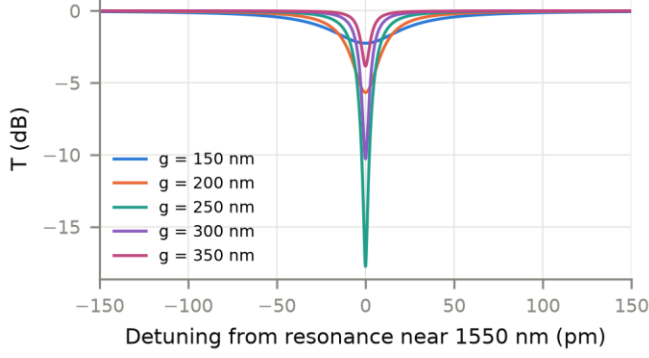


Fig. 5. Expected ring through-port spectra, Eq. (6).

ebeam_dc_halfkring_straight, $w = 500$ nm, $R = 10$ μ m, $L_c = 0$, $L = 58.3$ μ m, $n_g = 4.218$ (MODE). $\kappa^2(g)$: coupled-mode integral of the supermode splitting along the ring-bus gap (FEM, femwell [5]); agrees within 3 % with the PDK FDTD half-ring models at $g = 100$, 150 nm [3]. Round-trip loss $1 - a^2 = 0.0040$ vs. $\kappa^2 = 0.031$ – 0.001 : the gaps span over- to under-coupling (Table 3, Fig. 5). Expected $Q_i \approx 248$ k. Measured: fit Eq. (6) per resonance $\rightarrow r, a \rightarrow \alpha$; the gap series resolves the $r \leftrightarrow a$ ambiguity.

3.7 Loopback

$IL_{GC} = 1/2 T_{LB}$ [dB]. Expected (PDK TE GC model [3]): LBTE peak -4.6 dB at 1555 nm, 3 dB bandwidth 47 nm.

4 Fabrication

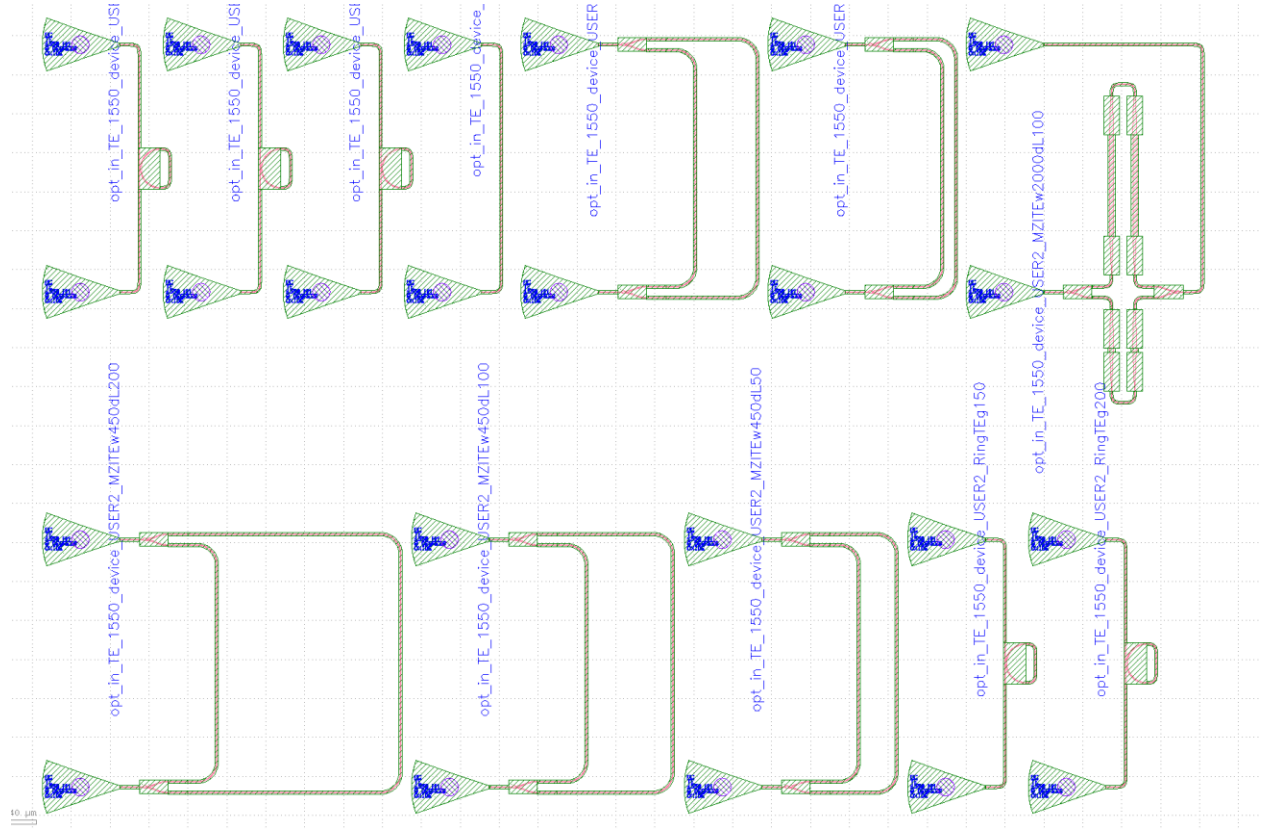


Fig. 6. Layout of EBeam_USER2 (605 × 410 μm).

TABLE 4. Devices on EBeam_USER2

Device	Pol.	Description
MZITEw450dL200	TE	MZI, Y-branch, $\Delta L=200 \mu\text{m}$, $w=450 \text{ nm}$
MZITEw450dL100	TE	MZI, Y-branch, $\Delta L=100 \mu\text{m}$, $w=450 \text{ nm}$
MZITEw450dL50	TE	MZI, Y-branch, $\Delta L=50 \mu\text{m}$, $w=450 \text{ nm}$
MZITEw450dL75	TE	MZI, Y-branch, $\Delta L=75 \mu\text{m}$, $w=450 \text{ nm}$
MZITEw450dL25	TE	MZI, Y-branch, $\Delta L=25 \mu\text{m}$, $w=450 \text{ nm}$
RingTEg150	TE	all-pass ring, $R=10 \mu\text{m}$, $g=150 \text{ nm}$
RingTEg200	TE	all-pass ring, $R=10 \mu\text{m}$, $g=200 \text{ nm}$
RingTEg250	TE	all-pass ring, $R=10 \mu\text{m}$, $g=250 \text{ nm}$
RingTEg300	TE	all-pass ring, $R=10 \mu\text{m}$, $g=300 \text{ nm}$
RingTEg350	TE	all-pass ring, $R=10 \mu\text{m}$, $g=350 \text{ nm}$
LBTE	TE	GC loopback, $L=144 \mu\text{m}$
MZITEw2000dL100	TE	MZI, Y-branch, $\Delta L=100 \mu\text{m}$ in $2 \mu\text{m}$ arms

KLayout, SiEPIC-Tools [6], EBeam PDK [3] (Fig. 6, Table 4). TE GCs, 127 μm pitch; labels opt_in_TE_1550_device_USER2_<name>. Functional check: 0 errors; DRC: 0 errors. Electron-beam lithography, Applied Nanotools NanoSOI [7], SiEPIC openEBL.

5 Experimental Data

To be completed (1500–1600 nm sweep, 1 pm step).

6 Analysis

To be completed.

7 Conclusion

To be completed.

References

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